

Subject: 23CSE301

Lab Session: 05

Notes:

1. Please read the assignment notes carefully and comply to the guidelines provided.
2. Code should be checked into the GitHub. These details shall be provided in the Lab.

Coding Instructions:

1. The code should be modularized; The asked functionality should be available as a function. Please create multiple functions if needed. However, all functions should be present within a single code block, if you are using Jupyter or Colab notebooks.
2. There should be no print statement within the function. All print statements should be in the main program.
3. Please use proper naming of variables.
4. For lists, strings and matrices, you may use your input values as appropriate.
5. Please make inline documentation / comments as needed within the code blocks.

6. DO NOT USE GenAI tools for coding work for this lab session.

Main Section (Mandatory):

Please use the data associated with your own project. This assignment deals with classification models.

<https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>

Please use help manuals of sklearn package to gain understanding of the model behaviors as well as ways to use various package functionalities.

Project teams who have not completed last week's assignments, please complete them.

A1. Create a modular design for implementing the k-NearestNeighbors classifier. The modules involved should consider the following functionalities. Reuse existing codes already developed for earlier lab sessions.

- a. **Encoding:** Convert data if categorical
- b. **Data Imputation:** Missing values of data to be filled in with appropriate central tendencies (Mean, Median & Mode)
- c. **Distance Calculation:** based on selected mode, distance may be calculated using any metrics.
- d. **Sorting:** implement sorting algorithms; implement at least 3 different sorting algorithms learnt in DSA subject. The sorting algorithm should be made available as a config parameter.
- e. **Identify Neighbors:** identify k-nearest neighbors based on distance; make sure to implement tie breaking mechanism for equal distance.
- f. **Class Evaluation and Assignment:** Identify the class label based on majority voting principle. Assign the class label to the test pattern. Make sure to implement the tie breaking mechanism.

A2. Modify appropriate modules to implement weighted kNN classification algorithm.

A3. Divide the dataset in your project into two parts – train & test set. To accomplish this, use the `train-test_split()` function available in SciKit. See below sample code for help:

```
>>> import numpy as np

>>> from sklearn.model_selection import train_test_split

>>> X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3)
```

X is the feature vector set for your project and y is the class levels for vectors present in X.

Note: Before set split, make sure you have only two classes. If your project deals with multi-class problem, take any two classes from them.

A4. Train a kNN classifier (k =3) using the training set obtained from above exercise. Following code for help:

```
>>> import numpy as np

>>> from sklearn.neighbors import KNeighborsClassifier

>>> neigh = KNeighborsClassifier(n_neighbors=3)

>>> neigh.fit(X_train, y_train)
```

A5. Test the accuracy of the kNN using the test set obtained from above exercise. Following code for help

```
>>> neigh.score(X_test, y_test)
```

A6. Use the predict() function to study the prediction behavior of the classifier for test vectors.

```
>>> neigh.predict(X_test)
```

A7. Implement the above-mentioned functions in your developed package. Following functions should be developed.

- a. Fit()
- b. Predict()
- c. Score()

A8. Make a comparative analysis between the function developed by you and the package function usage. Perform this experiment for a range of values for k. Make a plot of accuracies obtained from your developed function and the package function.

A9. Repeat the classification experiments with weightedkNN function developed by you. Make a comparison with A8 results.

Report Assignment:

1. Update your understanding of your project in the introduction section of the report.
2. Study the downloaded papers & update the literature survey section of your report.
3. Expand the methodology and results sections with outcomes of this experiments & results obtained. Please discuss your observations, inferences in results & discussion section. Please conclude the report appropriately with these experiments. Consider following points for observation analysis & inferences.
 - Do you think the classes you have in your dataset are well separated? Justify your answer.

- Explain the behavior of the kNN classifier with increase in value of k . Explain the scenarios of over-fitting and under-fitting in kNN classifier.
- Do you think the kNN classifier is a good classifier based on the results obtained on various metrics?
- Do you think the model has regular fit situation? Use train and test set performances to arrive at this inference.
- When do you think a situation of overfit happens for kNN classifier?